

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 09 · LECTURE NOTES

Light Fields and Plenoptic Cameras

Capturing rays instead of pixels.

MODULE OUTCOMES

- Define the plenoptic function and the 4D light field.
- Explain how a plenoptic camera trades resolution for angle.
- Explain digital refocusing and synthetic aperture.
- Explain why light field cameras failed commercially.
- Identify where light fields are actually used.

1 The plenoptic function

$$P(x, y, z, \theta, \phi, \lambda, t)$$

The **plenoptic function** describes the radiance at every point in space, in every direction, at every wavelength, at every instant. It is everything that could possibly be seen from anywhere — seven dimensions, and far too much to record.

$$L(u, v, s, t)$$

RADIANCE INVARIANCE REDUCES SEVEN DIMENSIONS TO FOUR

Fix the time and the wavelength and you have five dimensions: three of position and two of direction.

Now recall CSCE 647 Module 01: radiance is invariant along a ray in free space. So specifying a point and a direction over-determines the ray — every point along that ray carries the same radiance, so they are not independent measurements.

A ray in free space is therefore determined by where it crosses two parallel planes — two coordinates on each, four in total. This is the **two-plane parameterisation** and the resulting $L(u,v,s,t)$ is the **4D light field**.

The same physical fact that made ray tracing a lookup rather than an integral makes light field capture finite rather than impossible. It is worth noticing that the two courses depend on the same property for opposite purposes.

A PHOTOGRAPH IS A 2D SLICE OF A 4D FUNCTION

A conventional camera integrates over its aperture. Every ray arriving at a given sensor location, from every direction the aperture admits, is summed into a single value.

That integration is irreversible. Once the rays have been added together, there is no way to recover which came from which direction — the angular information is destroyed at the moment of capture, before any processing.

A light field camera records the rays separately, retaining the direction each arrived from.

So the integration can be performed afterwards, and differently. A different aperture, a different focal plane, a slightly different viewpoint — all are different integrals over the same recorded data. **Capture the function; compute the photograph.** This is Module 06's closing idea in its purest form.

2 Capturing a light field

Method	Mechanism	Trade-off
Camera array	Many cameras arranged on a plane, each recording one angular sample.	Full spatial resolution per view , and physically large, expensive, and requiring careful calibration. The Stanford array had 128 cameras.
Gantry	A single camera moved mechanically to each position in turn.	Cheap and high quality, and static scenes only — capture takes minutes.
Plenoptic (microlens array)	A microlens array placed just in front of the sensor; the pixels beneath each microlens record different directions.	A single shot, from a hand-held camera — and the spatial resolution falls catastrophically. See below.
Coded mask	A patterned attenuating mask near the sensor; the light field is reconstructed computationally from the multiplexed measurement.	Compressive — fewer measurements than the full light field, recovered by assuming sparsity. Requires a reconstruction step and loses light to the mask.

THE RESOLUTION TRADE IS BRUTAL, AND IT DECIDED THE MARKET

A microlens array places a small lens over each group of sensor pixels. Each microlens forms a tiny image of the main lens's aperture, so the pixels beneath it record light arriving from different *directions* through that aperture.

So each microlens contributes one spatial sample and as many angular samples as it covers pixels. The sensor's resolution is divided between the two.

A 20-megapixel sensor with 10×10 angular sampling yields a 0.2-megapixel image. Two hundred thousand pixels from a twenty-million-pixel sensor — and consumers compare megapixels, directly and unforgivingly.

Later designs recovered some of it by placing the microlens array at a different position so that it images the focal plane rather than the aperture — trading angular range for spatial resolution instead. **The trade moved; it did not disappear**, because the sensor has a fixed number of pixels and four dimensions to cover.

3 What you can compute

- **Digital refocusing.** Integrate over a *sheared* slice of the 4D function; different shears correspond to different focal planes. See below.
- **Synthetic aperture.** Integrate over a chosen subset of the angular samples. A small central subset gives a small effective aperture and deep depth of field; the full set gives the widest aperture the capture supports. **Aperture becomes a post-processing parameter.**
- **Viewpoint change,** within the extent of the capture aperture. The parallax available is small — limited by the physical size of the lens or the array — but it is genuine and free.
- **Depth estimation.** The focal setting at which a region appears sharpest indicates its depth, so a **depth map comes from a single exposure** with no stereo baseline and no projected pattern.
- **Seeing through partial occluders.** A synthetic aperture wider than an occluding object integrates light arriving around it from many directions, so the occluder blurs into near-transparency while the subject behind stays sharp. **Looking through foliage with a camera array is the most striking demonstration of the idea** and is worth seeing.

REFOCUSING IS A SHEAR FOLLOWED BY AN INTEGRAL

Reparameterising the two planes to a different separation is a shear of the 4D light field. A ray's coordinates on the new planes are a linear function of its coordinates on the old ones, and the transformation is a shear in the (u,s) and (v,t) pairs.

Integrating the sheared function over the two angular dimensions produces an image focused at the depth corresponding to that shear.

So refocusing is: shear, then sum. A few lines of code once the light field is in hand, and it is the operation that made the idea famous.

And the Fourier slice theorem gives a faster route. A 2D slice through the origin of the 4D Fourier transform is the Fourier transform of the corresponding 2D projection — so refocusing can be performed by extracting a slice in frequency space and inverse transforming, at $O(n^2 \log n)$ rather than $O(n^4)$. The same theorem underlies computed tomography, which is another place projections and slices meet.

4 What actually happened

LYTRO FAILED, AND THE REASONS ARE WORTH UNDERSTANDING

Resolution. The first consumer Lytro produced roughly a one-megapixel final image at a time when phones produced twelve. **The headline feature cost exactly the quantity consumers use to compare cameras**, and no amount of explanation overcame that.

The problem it solved was not acute. Autofocus was already good and improving; photographers did not frequently find themselves wishing they could change the focus after the fact. **A technically impressive solution to a problem few people had** is a recurring pattern worth recognising.

Depth estimation arrived by other routes. Dual-pixel sensors, stereo from two lenses, and learned monocular depth all gave adequate depth maps without surrendering any resolution at all.

And bokeh could be synthesised (Module 11) convincingly enough for the intended purpose. **The capability survived; the capture method did not** — portrait mode is what people wanted from light field photography, and it was delivered without it.

Domain	Use	Why it works there
Microscopy	Single-shot volumetric capture of fast biological processes — neural activity, beating cilia.	The subject moves faster than a focal sweep allows , and microscope resolution is abundant relative to the field of view. The trade that killed the consumer camera is acceptable here.
Industrial inspection	Depth and surface detail from one exposure, on a production line.	Robustness and single-shot capture matter more than resolution.
Film and VFX	Capture rigs for relighting, view synthesis, and set reconstruction.	Cost and bulk are acceptable; the output is a 3D asset rather than a photograph.
Display research	Light field displays present a genuine range of focal depths.	This would resolve the vergence-accommodation conflict of CSCE 650 Module 02, which no shipping headset solves.
Neural scene representations	NeRF and its successors.	See below.

NEURAL RADIANCE FIELDS ARE THE SAME IDEA, REACHED DIFFERENTLY

A NeRF represents a scene as a function from position and viewing direction to radiance and density — which is the plenoptic function of §1, **learned rather than sampled**.

Instead of capturing the 4D function densely, which was always the impractical part, a network is fitted to a sparse set of ordinary photographs and can then be queried for any ray.

This dissolves the sampling problem that defeated light field capture. You no longer need to record every ray — only enough photographs to constrain the fit, which is a few dozen rather than millions of samples.

And it inherits Module 13's question in full. A novel view rendered from a NeRF is *inferred* from the training photographs, not measured — it is a plausible completion, and in regions the training views did not cover it is a confident invention. **The distinction between measurement and inference, which this course keeps returning to, applies here as sharply as anywhere.**

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Levoy & Hanrahan — Light Field Rendering (1996, free)	The paper that introduced the 4D parameterisation of §1. Short and foundational.
Ng — Digital Light Field Photography (thesis, free)	The plenoptic camera and the refocusing mathematics of §3, by Lytro's founder. The Fourier slice refocusing is here.
Stanford Light Field Archive (free data)	Captured light fields to experiment with, including the camera array datasets. You can implement refocusing on real data without any hardware.
Mildenhall et al. — NeRF (free)	The §4 connection. Read it as a light field paper and the lineage is obvious.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Download a light field from the Stanford archive and visualise slices of the 4D function in several ways.
- 2 Implement digital refocusing by shear-and-sum. Produce a focal stack from a single light field.
- 3 Implement synthetic aperture and produce the same scene with deep and shallow depth of field.
- 4 Implement refocusing via the Fourier slice theorem and compare the cost against shear-and-sum.

- 5 Estimate depth from the light field by finding the focal setting that maximises local sharpness. Compare against the dataset's ground truth if available.
- 6 Use a camera array dataset to see through a partial occluder. This is the most striking demonstration in the module.
- 7 Compute the spatial resolution a plenoptic camera would give for a sensor and angular sampling of your choice, and compare against a conventional camera with the same sensor.
- 8 Synthesise a light field from a rendered scene using CSCE 647's renderer, and verify your refocusing against ground-truth renders at each focal depth.
- 9 Write a page on why Lytro failed, drawing on §4, and identify another technology that followed the same pattern.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the plenoptic function and explain the reduction to four dimensions.
- 2 Why is a conventional photograph a 2D slice, and what is irreversibly lost?
- 3 Name four capture methods with their trade-offs.
- 4 Compute the output resolution of a plenoptic camera from sensor resolution and angular sampling.
- 5 Name five operations computable from a light field.
- 6 Describe refocusing as an operation on the 4D function, and the faster frequency-domain route.
- 7 Give four reasons the consumer light field camera failed.
- 8 Name four places light fields are genuinely used.
- 9 What is a NeRF in terms of this module, and what question does it inherit?